Counterfactual Decision Stability for High-Stakes Mandarin ASR: Evaluating Safety-Relevant Instability Beyond Transcript Accuracy

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Abstract

Speech-driven decision systems increasingly use automatic speech recognition (ASR) outputs to support routing, compliance monitoring, escalation, and abstention. In high-stakes Mandarin anti-fraud calls, a small ambiguity in negation, amount, actor, action, time, intent, or scam pattern may leave a transcript close under conventional similarity metrics while changing the safe downstream action. Current ASR evaluation remains primarily transcript- or semantic-similarity centered and does not directly test whether decisions are stable under plausible ASR alternatives.

We propose Counterfactual Decision-Stability ASR (CDS-ASR), a consequence-centered evaluation framework for high-stakes speech-to-decision workflows. CDS-ASR extracts decision-critical risk atoms, generates acoustically and semantically plausible transcript variants, maps each transcript or variant to a declared triage action, and scores decision instability using Counterfactual Escalation Instability Score (CEIS). We evaluate the framework in a Mandarin anti-fraud setting using a six-model 258-row ASR split comparison, a selected-300 high-stakes provenance surface, and a completed 300-row dual-reviewer audit covering 900 model-level assessments per reviewer.

The audit shows strong inter-reviewer reliability across decision-change, semantic-risk, expected-safe-action, and confidence labels, with Cohen's kappa values of 0.850, 1.000, 0.851, and 0.934, respectively. Final aggregate regeneration finds 14 row-level decision-change positives and 6 row-level severe missed-escalation positives. Because the severe-positive count is below the pre-specified primary-endpoint threshold, severe-miss recovery is reported as descriptive high-severity replay, while decision-change prediction becomes the empirical endpoint. WER and CER remain necessary surface baselines, SRES remains the strongest thresholded semantic-risk baseline, and CEIS provides a scoped decision-stability layer and fusion input. The results support CDS-ASR as a privacy-preserving companion to transcript accuracy for high-stakes ASR evaluation.

Introduction

Speech is no longer used only as an archival record of what happened in a call. In modern contact-center systems, spoken interaction is increasingly converted into operational evidence: transcripts, summaries, categories, compliance cues, real-time alerts, routing signals, escalation support, and machine-generated assistance. Commercial contact-center analytics already describe voice conversations and transcripts as inputs for automated categorization, summarization, compliance monitoring, and real-time assistance (Amazon Web Services 2026a, 2026b). This shift changes the role of automatic speech recognition (ASR). A transcript is not merely an intermediate text artifact; it can become a decision substrate that influences whether a case is routed to routine handling, priority review, conservative escalation, or no further action.

This speech-to-decision setting is especially consequential in anti-fraud calls. Taiwan's National Police Agency describes the 165 hotline as a public anti-fraud channel where staff record incident details and provide information to victims (National Police Agency, Ministry of the Interior 2024). The operational scale is large: the FBI's 2025 Internet Crime Report context reported more than one million IC3 complaints and large cyber-enabled fraud losses (Federal Bureau of Investigation 2026b, 2026a). Public anti-fraud hotlines and fraud-reporting channels receive calls in which small spoken details may carry large operational consequences: whether money has already been transferred, whether the amount is 30,000 or 300,000, whether the caller is certain or uncertain, who initiated contact, when the event occurred, and whether the scenario matches a known scam pattern. These details are often short lexical spans, but they are not minor transcription errors. They are decision atoms. If an ASR system misrecognizes or ambiguously represents such atoms, the downstream system may change the expected safe action even when the overall transcript remains close to a reference transcript.

Conventional ASR evaluation provides an essential starting point for this problem. Word error rate (WER) and character error rate (CER) make transcript quality auditable, comparable, and reproducible across models and datasets. They remain necessary surface metrics, especially for Mandarin ASR, where tokenization and character-level reporting policies must be explicit. However, transcript similarity alone does not describe the consequence of an error. Two transcripts can have similar WER or CER while differing on a negation, an amount, an actor, or a transfer action that changes the operational interpretation of the call.

Recent semantic and downstream-aware ASR metrics address part of this limitation. Semantic Distance and Aligned Semantic Distance move ASR evaluation beyond literal string matching by measuring whether recognition errors preserve meaning or align with downstream task behavior (Kim et al. 2021; Rugayan, Salvi, and Svendsen 2023). Confidence-aware transcript correction and post-hoc repair methods further improve the pipeline by attempting to correct likely

ASR errors while avoiding harmful changes to already accurate transcripts (Naderi et al. 2024). Selective prediction, reject-option classification, and conformal prediction also provide a broader methodological basis for abstention and uncertainty-aware action when prediction risk is high (Chow 1970; Geifman and El-Yaniv 2017, 2019; Angelopoulos and Bates 2021).

These methods make ASR systems more measurable, more semantic, more correctable, and more uncertainty-aware. The remaining gap is decision stability. A semantic metric can indicate whether two transcripts are close in meaning. A correction model can produce a more fluent or more likely transcript. A confidence filter can reduce unnecessary correction. Yet a high-stakes speech-driven workflow also needs to know whether another acoustically and semantically plausible transcript would change the downstream decision. In anti-fraud triage, the relevant safety question is therefore not only "How accurate is the transcript?" but also "Would a plausible ASR alternative lead to a different safe action?"

This paper proposes Counterfactual Decision-Stability ASR (CDS-ASR) to answer that question directly. CDS-ASR treats high-stakes ASR evaluation as a counterfactual decision-stability problem. Given an ASR transcript, the framework extracts decision-critical risk atoms, constructs acoustically and semantically plausible transcript alternatives around those atoms, maps the original and alternative transcripts to a declared downstream triage action, and scores the resulting instability using Counterfactual Escalation Instability Score (CEIS). The central idea is that ASR evaluation for high-stakes speech-to-decision systems should include whether the downstream decision remains stable under plausible transcript alternatives.

CDS-ASR is designed as a companion layer to transcript accuracy metrics. WER and CER remain necessary for surface-level model comparison. Semantic-risk scores remain important for meaning-aware evaluation. Confidence-aware correction remains useful for transcript repair. CDS-ASR adds a different evidence layer: it asks whether recognition uncertainty reaches a decision-critical atom and changes escalation, routing, conservative machine action, or abstention. This makes the proposed framework a scoped companion to ASR accuracy and semantic-risk evaluation.

We evaluate CDS-ASR in a Mandarin anti-fraud ASR setting. The evidence design separates four layers: a six-model 258-row split comparison for ASR model context, a selected-300 high-stakes provenance surface, a completed 300-row dual-reviewer audit, and 900 model-level assessments per reviewer. This separation is important because model-level hypotheses are clustered within audio rows and should not be treated as independent calls. The selected-300 audit surface is deliberately enriched for high-stakes decision evidence; it supports decision-stability analysis within the reviewed surface, with population-level fraud prevalence and deployment-wide risk estimation reserved for a separate validation layer.

The empirical endpoint is also scoped. The pre-specified primary endpoint concerned row-level severe missed-escalation positives under fixed-budget replay. Final aggregate regeneration found only six such positives, below the pre-specified threshold. The manuscript therefore uses the pre-declared failover endpoint: decision-change prediction under row-clustered analysis is treated as the primary empirical endpoint, while severe missed-escalation replay is reported as descriptive high-severity evidence. This claim-evidence alignment supports the main methodological contribution: measuring consequence-relevant instability beyond transcript similarity.

The contributions of this paper are fourfold. First, we formulate high-stakes ASR evaluation as a counterfactual decision-stability problem for speech-to-decision workflows. Second, we introduce CDS-ASR, including a risk-atom schema, plausible ASR alternative construction, a declared downstream decision function, and CEIS as a decision-instability score. Third, we provide a Mandarin anti-fraud evaluation with six-model ASR comparison, selected high-stakes provenance, completed dual-reviewer human audit, and aggregate-only reproducibility boundaries for sensitive speech evidence. Fourth, we show how decision-stability evidence can support conservative action, abstention, and reviewer-visible governance within a human-review workflow.

By shifting the evaluation question from transcript similarity alone to decision stability under plausible ASR alternatives, CDS-ASR provides a consequence-centered companion layer for high-stakes ASR. The proposed framework is designed for ASR safety audit, risk-aware routing, manual-review prioritization, conservative escalation, and machine abstention. Its operating scope is staff-review intake support and aggregate evaluation, with production governance and adverse-action settings reserved for separate validation and human-review controls.

Related Work

Transcript-Centered ASR Metrics

Transcript accuracy metrics provide the reproducible baseline for ASR comparison. WER and CER remain necessary because they allow stable comparison across models and splits when tokenization, normalization, micro/macro scope, and zero-reference handling are declared. This manuscript keeps WER/CER as auditable surface metrics and uses the repo's metric audit to define the Chinese ASR reporting policy. CDS-ASR builds on that surface layer by asking which transcript differences reach decision atoms and change safe action.

Semantic And Downstream-Aware ASR Evaluation

Semantic ASR metrics extend transcript evaluation toward meaning preservation and downstream task behavior. Kim et al. introduce Semantic Distance for ASR performance analysis toward spoken-language understanding, motivating the move from literal correctness to semantic correctness for intent recognition, slot filling, semantic parsing, and named entity recognition (Kim et al. 2021). Rugayan et al. evaluate Aligned Semantic Distance against WER and show its value for human perception and downstream NLP tasks (Rugayan, Salvi, and Svendsen 2023). CDS-ASR uses this line of work as a foundation and adds a direct decision-stability target: whether plausible ASR alternatives change a high-stakes downstream decision.

Transcript Repair And Conservative Decision Action

LLM and confidence-aware correction methods provide transcript-repair baselines. Naderi et al. show that confidence-based filtering can guide post-hoc LLM correction of ASR transcripts and reduce the chance of changing likely accurate transcripts (Naderi et al. 2024). CDS-ASR complements this line by evaluating the decision interval that remains when plausible ASR alternatives affect risk atoms. Recovery in this manuscript is automatic and machine-bounded: human review supplies evaluation labels and governance evidence, while risk-triggered policies supply the scoped intervention test.

The conservative-action layer also connects to selective prediction and reject-option research. Chow formalizes the error-reject tradeoff for recognition systems (Chow 1970), while modern selective classification and SelectiveNet show how neural systems can abstain or reject to manage risk-coverage tradeoffs (Geifman and El-Yaniv 2017, 2019). Conformal prediction further motivates distribution-free uncertainty sets for high-risk systems (Angelopoulos and Bates 2021). CDS-ASR applies this governance intuition to speech-driven decision evidence: when transcript alternatives affect decision atoms, the system should expose the instability or take conservative action rather than treat one transcript as fully stable.

High-stakes speech domains reinforce the need for consequence-aware ASR evaluation. In psychotherapy, Miner et al. show that ASR feasibility depends on the clinical use case and that individual-level safety monitoring requires a more cautious threshold than population-level language analysis (Miner et al. 2020). CDS-ASR brings the same claim-evidence discipline to anti-fraud calls by separating transcript accuracy, semantic risk, decision instability, and aggregate recovery evidence.

This paper positions ASR models as evidence-producing components. The protagonist is the downstream decision that changes when an acoustically plausible transcript difference lands on a decision-critical atom.

Method

Problem Formulation

We study speech-driven decision systems where an audio segment is transcribed by ASR and then used by a downstream escalation workflow. The downstream label space includes routine review and higher-priority escalation states. The primary unit is the audio row / case. Model-level hypotheses are secondary observations clustered within the row and are used to form row-level scores by taking the maximum score over eligible model hypotheses.

The paper asks whether the decision remains stable under plausible transcript alternatives. This makes transcript accuracy a supporting layer and decision-stability the primary high-stakes safety target.

CDS-ASR Pipeline

CDS-ASR evaluates high-stakes ASR through a decision-stability pipeline:

```
audio
-> ASR transcript and runtime/quality signals
-> risk atom extraction
-> plausible counterfactual transcript variants
-> downstream decision model
-> SRES and CEIS scoring
-> constrained recovery or conservative machine action
```

Risk atoms are transcript spans whose alternatives can affect downstream decisions. This manuscript focuses on negation, amount, action, actor, time, intent, uncertainty, and scam-pattern atoms.

CEIS scores the maximum decision-flip risk among plausible variants. In this paper-facing implementation, the plausibility term is a bounded proxy rather than a calibrated acoustic posterior:

$$\mathrm{CEIS}(x) = \max_{v \in V(x)} \Big[P(v \mid x) \cdot w_{\mathrm{atom}}(v) \cdot d_f\big(f(x), f(v)\big) \Big].$$

$P(v \mid x)$ denotes a bounded proxy plausibility score derived from model disagreement, Mandarin phonetic ambiguity, domain-slot alternatives, and available ASR/runtime signals. $w_{\mathrm{atom}}(v)$ is a risk-atom class weight defined by the decision-critical atom schema. $d_f(f(x), f(v))$ maps the downstream label space into an ordinal safety distance over `no_escalation`, `review`, `priority_review`, `critical_escalation`, conservative machine action, and abstention, with critical misses treated as the highest-risk direction for conservative action.

Downstream Decision Function

The downstream decision function `f` is a declared retrospective policy abstraction for the final aggregate regeneration. It maps each transcript or plausible variant into a declared triage action:

$$
\begin{aligned}
f(t) \in \mathcal{A} = \{&\text{no escalation}, \text{manual review}, \text{priority review}, \text{critical escalation},\\
&\text{conservative machine action}, \text{abstain}\}.
\end{aligned}
$$

The distance function is policy-aligned rather than language-model-generated: same-action pairs have distance 0, neighboring review/escalation states have distance 1, and movement from no escalation or routine handling to critical escalation receives the largest distance. Unsafe downrouting of a critical event receives the maximum penalty used by the policy replay. The method contract is versioned in `80_semantic_risk_asr/paper/final_csl_audit_package_2026_06_03/downstream_decision_contract.md` and `80_semantic_risk_asr/paper/final_csl_audit_package_2026_06_03/ceis_config.json` so CEIS rankings can be reconstructed from aggregate artifacts.

CEIS Scale Discipline

All CEIS components are treated as bounded comparison terms. The $P(v \mid x)$ term represents proxy plausibility derived from model disagreement, Mandarin phonetic ambiguity, domain-slot alternatives, and available runtime signals. $w_{\mathrm{atom}}(v)$ and $d_f(f(x), f(v))$ provide the risk-atom and decision-distance handles that make ablation interpretable. The submission hardening plan therefore treats ablation as a direct contribution test: CEIS full, CEIS without atom weights, CEIS without plausibility, CEIS binary atom, and policy-distance-only. The final ablation suite shows that policy-distance-only matches full CEIS on the aggregate reviewed surface. The final manuscript therefore treats plausibility and atom weights as explicit method components, calibration handles, and interpretability/localization layers rather than empirically proven performance drivers on this selected surface.

Thresholds reported in the current predictor table are diagnostic operating points selected on the scoped reviewed audit. They are useful for aggregate comparison and reviewer inspection, but they are not frozen deployment thresholds unless later selected on a separate development set.

Recovery remains automatic and machine-bounded. Human review is used as an evaluation and governance layer, not as the proposed recovery method. The recovery policies compare no recovery, confidence-only trigger, SRES-triggered recovery, CEIS-triggered conservative action, and CEIS ensemble arbitration.

Risk Atom Schema

The risk atom schema focuses on decision-critical spans:

Risk atom	Decision role	Example effect
Negation	reverses event state	transfer vs no transfer
Amount	changes severity	30,000 vs 300,000
Action	changes case stage	asking vs reporting vs transferring
Actor	changes scam interpretation	bank, police, family, customer service
Time	changes urgency	today, yesterday, next week
Intent	changes routing	inquiry, report, complaint
Uncertainty	changes confidence and safe action	sure, maybe, not sure
Scam pattern	changes case type	investment, fake police, installment cancellation

Counterfactual Variant Contract

Counterfactual variants are plausible ASR alternatives concentrated around risk atoms. They are not generic paraphrases. A variant can be supported by acoustic ambiguity, Mandarin phonetic confusion, domain-slot alternatives, or model disagreement. The paper-facing claim is not that every possible variant is enumerated; it is that the generated alternatives make downstream decision-instability measurable.

Recovery Policy Contract

The five-policy recovery experiment is:

1. no recovery;
2. confidence-only trigger;
3. SRES-triggered recovery;
4. CEIS-triggered conservative action;
5. CEIS ensemble arbitration.

The scoped recovery claim is evaluated on aggregate human-reviewed selected-300 evidence. Per-row transcript-bearing details stay local. At the policy layer, both SRES-triggered recovery and CEIS-triggered conservative action are reported as risk-triggered conservative policies; CEIS's distinct contribution is evaluated primarily in the predictor layer, the decision-stability framing, and the planned ablation suite over atom weights, plausibility, and binary atom variants.

Reproducibility Layer

The paper-facing ASR surface metric is `cer_zh_micro`. The supplemental word metric is `wer_zh_jieba_micro`. Raw whitespace WER and legacy stored WER fields are audit-only because they are unstable for unsegmented Chinese transcripts.

The metric policy is validated by `70_experiments/runs/wer_metric_audit_2026_05_25/`, which records manifest checks, zero-reference-unit checks, tokenizer and normalization policy, and `jiwer` cross-checks.

The target transcription locale is Taiwan Traditional Chinese. New candidate models must satisfy the strict Taiwan Traditional Chinese locale gate before promotion. Previously completed comparable baselines are retained only as disclosed baselines, with locale-violation counts reported and no promotion claim attached. Post-decode OpenCC or other conversion can be evaluated only as a deployment repair lane; it cannot be used to claim that the raw ASR model passed the locale gate.

Selected-300 human audit evidence is tracked through aggregate outputs. The local transcript-bearing reviewer sheets, merged sheets, and completed package remain outside the committed release boundary. The reviewed scope is risk atoms, decision-change labels, expected safe action, confidence, per-model assessment fields, and per-row timing. The current aggregate status is dual-reviewer complete: reviewer 1 and reviewer 2 each completed 300/300 row-level records and 900/900 model-level assessments. Cohen's kappa is 0.849970 for decision-change, 1.000000 for semantic-risk label, 0.851426 for expected safe action, and 0.934274 for annotation confidence.

All paper-facing claims point to aggregate artifacts. Raw audio, raw transcripts, selected row IDs, hypotheses, reviewer notes, and model weights remain governed local materials.

Evaluation Units And N-Ladder

The manuscript separates evidence units explicitly so that model assessments are not mistaken for independent audio rows.

Layer	Unit	N	Role
Test split	audio rows	258	ASR model comparison
Selected high-stakes provenance	selected candidate rows / outputs	300	row-selection and provenance
Human-reviewed audit rows	audio rows	300	dual-reviewer decision-critical review unit
Human-reviewed model assessments	model-row assessments	900 per reviewer	reviewer agreement and full selected-300 audit evidence

Because model assessments are clustered within selected audio rows, inferential uncertainty should be estimated with row-clustered bootstrap resampling or leave-one-row-out sensitivity analysis. The completed 300-row dual-reviewer audit strengthens the reliability layer; any predictor or recovery table regenerated from the full audit should keep the row-clustered design rather than treating model-level assessments as independent calls.

Statistical Inference And Operating-Point Selection

The primary statistical unit is the audio row. Model-level assessments are clustered within row and are secondary evidence. Final aggregate regeneration defines

$$s_i = \max_{h \in H_i} s_{ih},$$
$$y_i^{\text{severe}} = \mathbb{I}[\exists h \in H_i : y_{ih}^{\text{severe}} = 1] \, .$$

Here i indexes an audio row and H_i is the set of eligible model hypotheses for that row. Fixed-budget replay uses the selected-300 rows as the denominator, so 10%, 20%, 30%, and 40% budgets correspond to 30, 60, 90, and 120 triggered rows.

The final dual-reviewer decision-change label uses a deterministic two-reviewer consensus rule. Agreement cases map directly to `yes` or `no`; unresolved mixed cases remain `uncertain` and are reported through the yes-or-uncertain sensitivity target. This rule is reproducible and does not claim a separate adjudication panel.

The pre-specified primary endpoint was row-level severe-miss remaining at 10–20% fixed row budgets. The final 300/900 regeneration found only 6 row-level severe-miss positives, below the planned threshold of 20. The manuscript therefore activates the pre-declared failover: decision-change prediction under clustered analysis is the primary empirical endpoint, while severe missed-escalation replay remains descriptive high-severity evidence.

At fixed-budget cutoffs, tied score groups are reported with best-case and worst-case severe-miss remaining; claims use the worst-case boundary. Thresholds selected on the same reviewed audit surface are diagnostic operating points, not deployment thresholds. Selection sensitivity is reported as pattern-preserving, pattern-reversing, or underpowered; it is not presented as proof that enrichment or selection-circularity risk has been removed.

Selection Provenance And Enrichment

The selected-300 audit surface is an enriched high-stakes sample, not a prevalence-preserving sample of all anti-fraud calls. The selection summary uses aggregate-safe provenance signals from SRES scoring, CEIS scoring, and downstream escalation decisions, including high-risk missed, critical miss, unsafe down-routing, model disagreement, high proxy risk, risk-score fill, and clean-control strata. The aggregate selection thresholds recorded for the current package are `low_wer_threshold=10.0`, `sres_threshold=20.0`, and `ceis_threshold=5.0`.

Predictor results therefore support metric behavior on an enriched high-stakes audit surface rather than population-level risk prevalence. If a submission emphasizes CEIS/SRES separation, a sensitivity check should exclude rows directly selected by CEIS or SRES family signals and confirm whether the same directional pattern remains.

Variant Coverage And Human Review Reliability

The current aggregate-only coverage audit reports CEIS top-atom proxy coverage on the existing replay surface. It records aggregate proxy observations with top-atom counts of negation 47, amount 37, action 3, and actor 3. The completed selected-300 dual-reviewer audit provides the final submission validation layer, while the reviewed selection surface also covers risk-signal atoms in aggregate: action 25, actor 15, amount 23, negation 14, and scam pattern 23 selected rows. These counts support submission-safe coverage auditing, not release of variant text. Source-specific coverage is available for model-disagreement provenance in aggregate; phonetic, domain-slot, runtime-signal, rejected-variant, and full variant-generation logs are not currently reconstructed as release artifacts. Stronger generator claims therefore require a future aggregate variant-generation log.

Human labels now include a completed dual-reviewer audit across the full selected-300 surface. Both reviewers completed all row-level and model-level fields, required fields have no empty values, row-level `yes` decisions have at least one model-level `yes` or `uncertain`, critical atoms are contained in risk atoms, and `yes` rows have expected safe action other than `none`. The agreement layer records Cohen's kappa of 0.849970 for decision-change, 1.000000 for semantic-risk label, 0.851426 for expected safe action, and 0.934274 for confidence. This turns human audit reliability from a planned extension into a reviewer-visible contribution.

Experiments

Experiment 1 evaluates comparable ASR model evidence on the canonical 258-row test split. This table supports model-comparison and split evidence, not the paper-grade selected-300 risk/recovery claims.

Experiment 2 uses selected-300 high-stakes proxy outputs as input provenance. The selected-300 proxy outputs identify an enriched high-stakes evidence surface and the review sample, but their raw rows and transcript-bearing metric inputs remain local or ignored. This experiment supports selection provenance, not population prevalence.

Experiment 3 evaluates WER, CER, SRES, and CEIS against human-reviewed decision-change labels from the selected-300 audit. The latest human audit gate is complete at 300 rows with 900 model-level assessments and dual-reviewer agreement summaries. Final aggregate regeneration uses a deterministic two-reviewer consensus rule, treats model-level assessments as clustered within row, and uses decision-change prediction as the primary endpoint because row-level severe positives are sparse.

Experiment 4 evaluates five recovery policies against the same human-reviewed evidence layer. This is aggregate policy replay evidence for conservative action and descriptive high-severity replay; a live deployment trial is a later validation layer.

Candidate ASR and multimodal models are kept in a separate exploratory lane. They must not enter the main ASR table until they pass field-contract, runtime-validity, and Taiwan Traditional Chinese locale gates in order.

Results

The Results section separates comparable ASR evidence, candidate-lane boundaries, human-reviewed predictor evidence, and human-reviewed recovery evidence. This separation preserves claim-evidence alignment: the main ASR benchmark supports model-comparison context, while selected-300 human-reviewed outputs support paper-grade risk and recovery claims.

Human review supplies evaluation labels only. The proposed recovery policies are automatic, aggregate-evaluated policies; no transcript-bearing row content is required for paper-facing claims.

Main Results Tables And Figures

The main article uses a reduced figure and table set. The complete R-generated figure package remains available for reproducibility, but the body emphasizes method, core predictor evidence, risk mechanism, and policy replay.

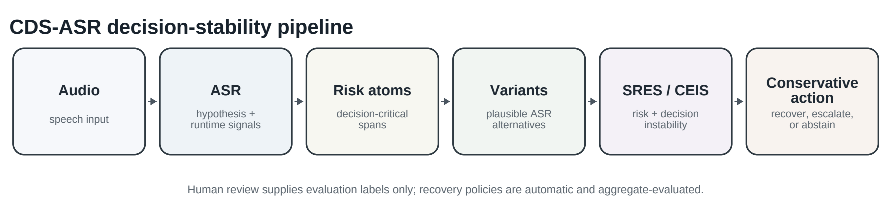

Figure 1: CDS-ASR converts audio into ASR hypotheses and runtime signals, extracts risk atoms, generates plausible variants, scores SRES/CEIS, and applies conservative action. Source: `method section`.

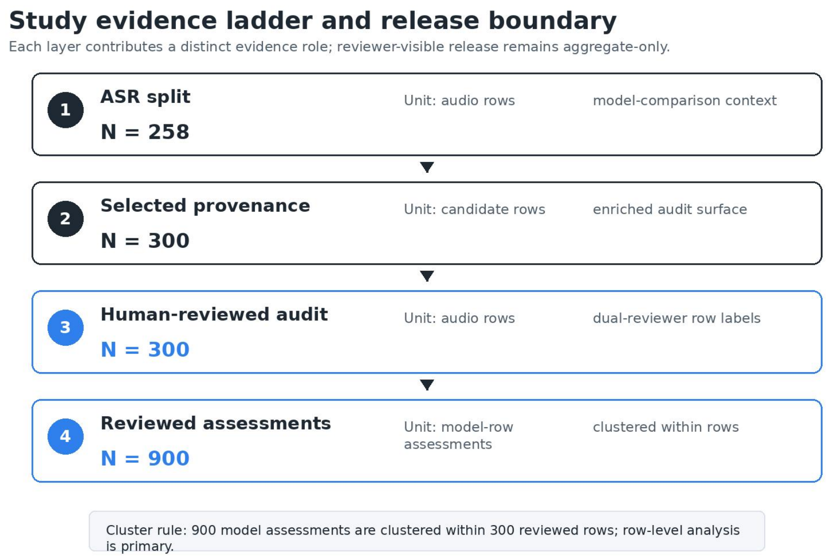

Figure 2: The study separates ASR split evidence, selected-300 provenance, the completed 300-row dual-reviewer audit, and 900 model-level assessments per reviewer. Source: `method evidence units and publishable evidence summary`.

Table 1 reports comparable 258-row split evidence. Run identifiers are kept out of the main table and remain in the aggregate source artifact. The table supports ASR model-comparison context only; final risk and recovery claims are tied to the selected-300 human-reviewed evidence.

Table 2 reports the final aggregate predictor comparison against human-reviewed `human_decision_change_yes` labels. WER/CER are transcript-surface baselines; SRES is the semantic-risk baseline; CEIS is the proposed decision-stability metric. The completed 300-row / 900-assessment regeneration is now recorded in `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/`. It reports 14 row-level decision-change positives, 20 yes-or-uncertain rows, and 6 row-level severe-miss positives. Because the severe count is below the pre-specified threshold of 20, the primary claim shifts to decision-change prediction and CEIS/SRES complementarity; severe-miss replay is descriptive.

This is the paper's central contribution pattern: CEIS operationalizes counterfactual decision stability, SRES remains the strongest semantic-risk baseline for several thresholded operating points, and the final claim is scoped to what the completed 300/900 evidence can support. A diagnostic residual-gain check fits

$$\operatorname{logit} \Pr(y_i = 1) = \beta_0 + \beta_1 \ \mathrm{SRES}_i,$$
$$\operatorname{logit} \Pr(y_i = 1) = \beta_0 + \beta_1 \ \mathrm{SRES}_i + \beta_2 \ \mathrm{CEIS}_i$$

Table 1: Main ASR benchmark on the canonical 258-row split. Run identifiers are reported in the supplement.

Model	Rows	CER zh micro	WER zh jieba micro	Unsafe downrouting	High-risk missed	Locale violations	Paper role
Breeze-ASR-25 partial encoder	258	15.04	21.53	7	4	0	Strongest ASR evidence layer
Breeze-ASR-25 LoRA	258	18.23	25.59	10	7	0	Split/model context
Breeze-ASR-25 base	258	22.72	30.39	34	30	0	Split/model context
Breeze-ASR-26	258	24.27	32.29	27	22	0	Split/model context
Whisper large-v2	258	24.72	32.23	33	28	1	Comparable baseline
Whisper small	258	34.86	43.44	76	70	4	Comparable baseline

CER zh micro is the primary ASR surface metric; WER zh jieba micro is supplemental. Unsafe downrouting and high-risk missed counts are split/model-comparison evidence, not final selected-300 human-reviewed risk claims.

Table 2: Final 300/900 predictor performance against deterministic dual-reviewer decision-change labels. Unit: model assessments clustered within 300 audio rows; primary endpoint uses the decision-change failover because row-level severe positives are 6.

Predictor	AUC	Threshold	Best F1	Precision	Recall	FN	Positives	Role
WER	0.811	42.590	0.253	0.189	0.385	16	26	Surface baseline
CER	0.832	37.500	0.321	0.300	0.346	17	26	Chinese surface baseline
SRES	0.720	270.000	0.453	0.444	0.462	14	26	Semantic-risk baseline
CEIS	0.718	5.000	0.407	0.364	0.462	14	26	Decision-stability layer
Fusion	0.720	0.783	0.453	0.444	0.462	14	26	Fusion baseline

Thresholds are diagnostic operating points selected on the scoped reviewed audit, not deployment thresholds. Fusion denotes the normalized maximum of SRES and CEIS. SRES has the strongest thresholded F1 in this table; CEIS is retained as a scoped decision-stability layer and fusion input rather than a total-superiority claim.

on the evaluable model-level consensus rows. The fused diagnostic model yields a small log-loss improvement with nearly unchanged AUC and F1, which supports complementarity rather than a total-superiority claim.

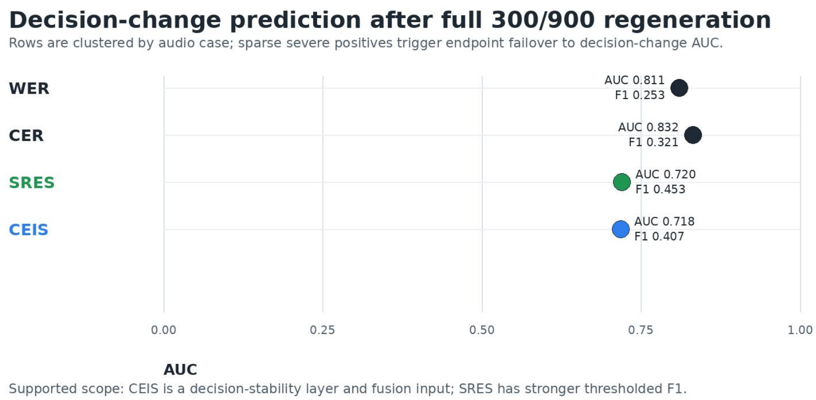

Figure 3: Final 300/900 regeneration evaluates decision-change prediction under clustered analysis; severe-miss replay is descriptive because row-level severe positives are sparse. Source: `final_csl_predictor_performance.tsv`; `final_csl_auc_delta_bootstrap.tsv`; `final_csl_residual_gain_after_sres.tsv`.

Final regeneration finds 6 row-level severe missed-escalation positives and 1 row-level critical miss. Because this positive count is below the pre-specified primary-endpoint threshold, recovery and fixed-budget frontiers are reported as descriptive high-severity replay rather than the primary empirical endpoint. Risk-triggered conservative policies remain useful for inspecting how many severe rows are captured at pre-specified row budgets, while residual unsafe downrouting remains a separate taxonomy and governance question.

A fixed-budget replay provides an additional operating view. In the final row-level regeneration, 10%, 20%, 30%, and 40% budgets correspond to 30, 60, 90, and 120 triggered rows. CEIS, SRES, and simple fusion policies all capture the sparse severe rows in the current aggregate replay, so the frontier supports descriptive stress testing and complementarity analysis rather than metric-replacement framing. Appendix Table A1 reports the row-level fixed-budget frontier from `final_csl_fixed_budget_frontier_row_level.tsv`.

Table 3: Row-level fixed-budget conservative replay on the final selected-300 audit surface.

Metric	Budget	Triggered	Severe rem.	Severe elim.	Unsafe rem.	Critical rem.	Tie worst	Role
SRES	10%	30	0	6	0	0	0	Semantic-risk replay
SRES	20%	60	0	6	0	0	0	Semantic-risk replay
CEIS	10%	30	0	6	0	0	0	CEIS replay
CEIS	20%	60	0	6	0	0	0	CEIS replay
Fusion	10%	30	0	6	0	0	0	Fusion replay
Fusion	20%	60	0	6	0	0	0	Fusion replay
Variant count	10%	30	5	1	16	1	6	Count stress test
Variant count	20%	60	2	4	8	0	5	Count stress test

The severe-miss baseline is 6 rows, so replay is descriptive high-severity evidence after the pre-declared failover to decision-change prediction. Variant-count-only is included as a stress test for max-over-variants behavior.

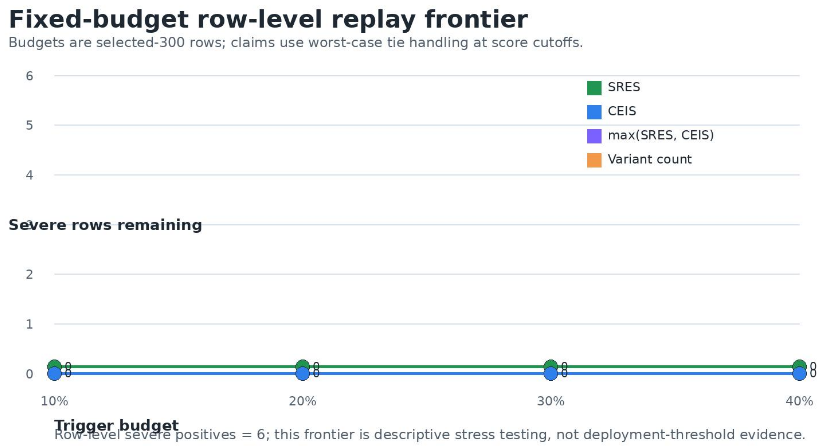

Figure 4: Fixed-budget policy replay reports row-level trigger budgets and tie-boundary ranges for descriptive severe-miss stress testing. Source: `final_csl_fixed_budget_frontier_row_level.tsv`.

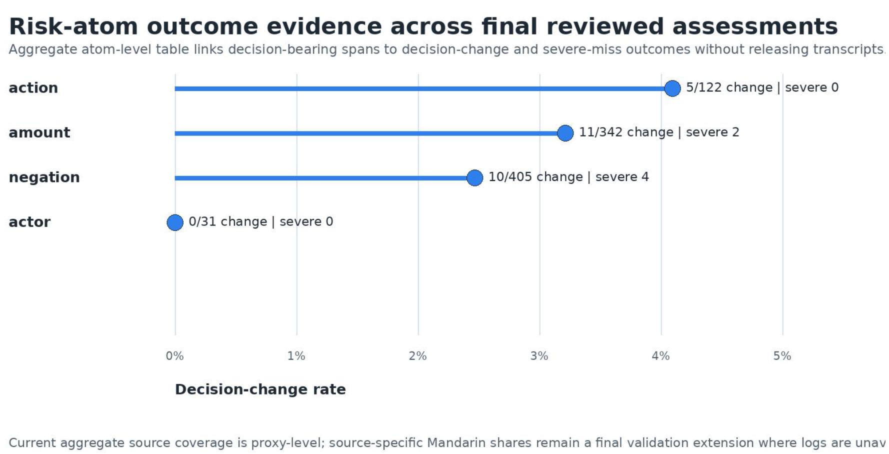

Figure 5: Human review connects risk-atom coverage to criticality and decision-change evidence across reviewed rows. Source: `human_audit_risk_atom_review.tsv`.

Word-level confusion entropy is a valuable next analysis, especially for frequently misrecognized decision words around negation, amount, transfer action, actor identity, and scam-pattern cues. The current submission package keeps that analysis out of the public figure set unless it is first converted into a redacted aggregate lexicon with no transcript spans, audio IDs, selected row IDs, hypotheses, or reviewer notes. A future local-only confusion lexicon can support a supplemental word-level entropy heatmap after privacy review.

Discussion

The evidence supports a consequence-centered ASR evaluation claim: correct WER/CER reporting is necessary, but transcript similarity alone does not capture high-stakes decision instability. The final selected-300 predictor table shows that transcript-surface metrics retain strong AUC on this scoped audit surface, while SRES remains the strongest thresholded F1 baseline and CEIS is retained as a counterfactual decision-stability layer and fusion input.

The model comparison should be presented as an evidence layer, not as a model leaderboard. The partial encoder is the strongest current ASR hypothesis generator on the comparable split, and the selected-300 human-reviewed evidence shows how downstream decision evidence refines transcript-centered model comparison.

The recovery result supports a scoped descriptive replay claim. Under the final row-level regeneration, severe missed-escalation positives are sparse, so risk-triggered conservative policies are used to inspect high-severity capture and residual unsafe categories rather than to carry the primary endpoint. CEIS is therefore best presented as a counterfactual decision-instability layer, while SRES remains a strong semantic-risk recovery and thresholding baseline.

Improving ASR remains necessary, but stronger transcript models do not replace decision-stability analysis. The paper studies residual instability after transcript scoring: whether plausible ASR alternatives around decision atoms would change escalation, routing, or conservative action. This makes CDS-ASR a safety layer for the remaining decision interval, not a substitute for ASR model improvement.

The aggregate-only artifact boundary is also a contribution to reproducible governance. It provides aggregate reproducibility, operation records, validation gates, dual-reviewer agreement summaries, and consistency checks under a privacy-preserving audit boundary.

The candidate lane is already bounded by evidence. Whisper large-v3, Whisper large-v3-turbo, SenseVoiceSmall, and Qwen3-ASR-0.6B have small-gate evidence but fail strict Taiwan Traditional Chinese locale promotion criteria. Qwen3-ASR-1.7B is stopped by fetch/load timeout, and Gemma 4 E2B/E4B require an isolated multimodal runtime. The next validation step is locale/runtime validation, not a broader full-split ASR run.

Supplementary Claim Registry

The claim registry is retained as supplementary governance evidence. It links paper-facing claims to aggregate artifacts, statistics, and claim-evidence alignment boundaries without taking space from the main result sequence.

Claim	Scope	Evidence artifact	Statistic	Limitation / defense
CEIS supplies a counterfactual decision-stability signal	Final 300-row / 900-assessment aggregate regeneration	`final_csl_predictor_performance.tsv`	decision-change prediction table	Scoped companion metric; SRES remains the strongest semantic-risk baseline for several thresholded views
Primary endpoint failover is active	Row-level severe positives are below the pre-specified threshold	`final_csl_row_level_positive_counts.tsv`	row-level severe positives 6	Severe frontier is descriptive high-severity evidence, not the primary endpoint
Fixed-budget replay is descriptive	Row-level selected-300 conservative replay	`final_csl_fixed_budget_frontier_row_level.tsv`	10–40% row budgets	Replay evidence, not live causal deployment or all-risk mitigation
Partial encoder is the strongest current ASR hypothesis generator	258-row split/model-comparison layer	`asr_cds_proxy_comparison.tsv`	`cer_zh_micro` 15.04, unsafe downrouting 7, high-risk missed 4	Split-level model-comparison context only
Selected-300 is a high-stakes audit surface	enriched selection provenance	`human_audit_selection_summary.json`, `selection_strata.tsv`	300 candidates, 300 reviewed rows, 900 model-level assessments per reviewer	Enriched audit surface; selection provenance disclosed
Dual-reviewer audit supports label reliability	completed selected-300 review	`human_audit_reviewer_agreement_summary.json`, `human_audit_reviewer_agreement.tsv`	Cohen's kappa 0.849970–1.000000 across key fields	Transcript-bearing reviewer sheets remain local
Aggregate-only release supports reviewer-visible auditability	release and operation-record layer	validation summaries, evidence matrices, operation records, figure scripts	gate state: roadmap complete, publishable ready, consistency 26/26	Public auditability with governed local transcript-bearing evidence

Ethics, Privacy, And Intended Use

This study treats transcript-bearing speech data as sensitive operational content. NIH human-subjects guidance treats the use, study, analysis, or generation of identifiable private information as a human-subjects trigger under the cited federal definition (National Institutes of Health 2024). This paper therefore keeps raw audio, transcript-bearing hypotheses, reviewer notes, row identifiers, local response sheets, and runtime logs outside the release boundary. Before any external data release or deployment claim, the institutional review, exemption, data-use, retention, encryption, and access control status must be documented.

CDS-ASR is intended for ASR safety audit, risk-aware routing, manual-review prioritization, conservative escalation, and machine abstention. It is not intended for automatic guilt or fraud determination, automatic account freezing, service denial, punitive action, or direct law-enforcement reporting without human review. "Conservative machine action" means preserving uncertainty, raising review priority, abstaining, or requiring human confirmation.

This intended-use boundary aligns the paper with risk-management governance rather than autonomous adverse decision-making. NIST describes the AI RMF as a voluntary framework for managing AI risks to individuals, organizations, and society and for incorporating trustworthiness considerations into AI design, development, use, and evaluation (National Institute of Standards and Technology 2026). For cross-border or future deployment contexts, the EU AI Act illustrates the broader regulatory direction toward risk-based obligations and high-risk AI governance (European Commission 2026). This manuscript contributes a scoped consequence-centered audit method and a privacy-preserving release boundary that can sit inside the appropriate institutional and legal review process.

Limitations And Threats To Validity

The human-reviewed evidence now covers the full selected-300 surface with two reviewers and 900 model-level assessments per reviewer. This supports a stronger high-stakes audit and reviewer-agreement claim while leaving population-level deployment prevalence to a later study.

Model-level assessments are clustered within selected audio rows and should not be treated as independent calls. The final 300/900 aggregate regeneration and CEIS ablation suite are incorporated in the present release-candidate. Remaining validation concerns are source-specific variant-generation coverage, population-prevalence estimation, and deployment-threshold calibration.

The number of positive decision-change cases remains a key calibration handle. The final CSL table should report the positive-label count under the completed dual-reviewer audit and preserve row-clustered uncertainty for AUC and threshold behavior.

The selected-300 audit surface is deliberately enriched for high-stakes signals.

It supports decision-stability behavior within the selected audit boundary, not population prevalence of ASR-induced harm across all calls.

The reported best thresholds are diagnostic operating points on the scoped reviewed audit. They can overstate deployment performance unless threshold selection is later frozen on a separate development set.

Human labels now include a completed dual-reviewer audit and reviewer-agreement summary. Cohen's kappa is 0.849970 for decision-change, 1.000000 for semantic-risk label, 0.851426 for expected safe action, and 0.934274 for annotation confidence. The present release-candidate incorporates final table regeneration and the CEIS ablation suite; the next validation layer is source-specific variant-generation coverage and deployment-threshold calibration.

Recovery evidence is aggregate-only. This protects transcript-bearing call and review content, but limits external row-level reproducibility.

CEIS depends on generated plausible variants and risk atom weights. Missed variants or misweighted atoms can affect instability scoring. The current aggregate package records risk-atom coverage and a variant-coverage status table, but a full source-specific aggregate variant generation audit remains a planned validation extension.

The Taiwan Traditional Chinese locale policy is strict by design. Candidate model rejection may reflect deployment-locale mismatch, not universal ASR inferiority.

Final row-level regeneration reports 6 severe missed-escalation rows and 22 unsafe-downrouting rows. The recovery frontier is therefore descriptive high-severity replay, and residual unsafe downrouting is reported through an aggregate breakdown rather than treated as handled. The evidence supports scoped decision-stability evaluation and conservative-action stress testing, not elimination of all safety risk.

The study evaluates whether decision-critical ASR risk can be detected and stress-tested under conservative-action replay within a deliberately enriched high-stakes audit surface. A deployment trial and population-prevalence study are natural next validation layers after the completed reviewer-agreement and ablation gates are incorporated.

Appendix / Artifact Availability

Artifact Availability Statement

The repository release includes aggregate run records, validation summaries, metric tables, evidence matrices, figure-generation code, and paper-facing documentation. Raw audio, raw transcripts, selected row identifiers, audio identifiers, model hypotheses, transcript-bearing runtime logs, reviewer response sheets, reviewer notes, and model weights are not released because they may contain sensitive call or review content.

Because transcript-bearing materials may contain sensitive call content, reproducibility is provided through aggregate validation artifacts, operation records, manifest checks, and consistency audits rather than through raw row release. This provides reviewer-visible auditability for model comparison, metric policy, selected-300 human-review status, predictor analysis, recovery-policy evaluation, candidate-lane boundaries, and consistency checks while preserving the local-only boundary for sensitive materials.

The reviewer package should include an aggregate manifest with artifact path, role, privacy class, generator, source inputs, SHA256, source git commit, timestamp, Python version, metric-library versions, tokenizer policy, locale normalization policy, decoding parameters where available, random seeds where used, and hardware summary. The manifest records auditability metadata without adding transcript-bearing row content.

Reviewer-Reproducible Aggregate Artifacts

- Artifact manifest: `80_semantic_risk_asr/paper/artifact_manifest.tsv`
- Artifact manifest generator: `80_semantic_risk_asr/paper/build_artifact_manifest.py`
- Experiment registry: `70_experiments/registry.tsv`
- Main 258-row comparison: `70_experiments/runs/janus_258_test_split_asr_cds_proxy/asr_cds_proxy_comparison.tsv`
- Metric-definition audit: `70_experiments/runs/wer_metric_audit_2026_05_25/journal_compliance_summary.json`
- **Final CSL evidence artifacts:** `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/`
- Final predictor table: `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/final_csl_predictor_performance.tsv`
- Final AUC delta bootstrap: `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/final_csl_auc_delta_bootstrap.tsv`
- Final SRES residual-gain diagnostic: `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/final_csl_residual_gain_after_sres.tsv`
- Final row-level positive counts: `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/final_csl_row_level_positive_counts.tsv`
- Final fixed-budget frontier: `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/final_csl_fixed_budget_frontier_row_level.tsv`
- Final residual unsafe breakdown: `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/final_csl_residual_unsafe_breakdown.tsv`
- Final CEIS ablation artifacts: `70_experiments/runs/janus_300_high_stakes_ceis_ablation_final_csl_2026_06_03/`
- Final CEIS ablation delta summary: `70_experiments/runs/janus_300_high_stakes_ceis_ablation_final_csl_2026_06_03/ceis_ablation_delta_summary.tsv`
- Completed 300-row dual-reviewer agreement summary: `70_experiments/runs/janus_300_high_stakes_human_audit_completed_300_dual_reviewer_2026_06_02/human_audit_re`

viewer_agreement_summary.json
- Completed 300-row dual-reviewer agreement table: `70_experiments/runs/janus_300_high_stakes_human_audit_completed_300_dual_reviewer_2026_06_02/human_audit_reviewer_agreement.tsv`
- **Legacy / provenance artifacts:** selected-300 human-audit refresh: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_refresh_summary.json`
- Legacy predictor comparison, retained as pre-final provenance: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_comparison.tsv`
- Legacy recovery replay, retained as pre-final provenance: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison.tsv`
- Claim registry: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/claim_registry.tsv`
- CEIS method summary: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/ceis_method_summary.tsv`
- Selection provenance summary: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/selection_provenance_summary.tsv`
- Counterfactual variant coverage status: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/counterfactual_variant_coverage_summary.tsv`
- Post-review evidence checklist: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_post_review_evidence_summary.json`
- Publishable evidence audit: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/publishable_evidence_completion_summary.json`
- Consequence evidence matrix: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/consequence_evidence_matrix.tsv`
- Roadmap completion audit: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/postdoc_roadmap_completion_summary.json`
- Evidence-chain consistency audit: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/evidence_chain_consistency_summary.json`
- Candidate bounded recheck: `70_experiments/runs/asr_candidate_current_recheck_2026_05_26/summary.json`

Supplementary Candidate And Mechanism Figures

The candidate lane and aggregate mechanism figures are retained as supplementary evidence. They support reproducibility, scope tracing, and mechanism inspection while keeping the main article focused on method, core predictor evidence, risk mechanism, and policy replay.

Table 4: Candidate and exploratory lane evidence.

Candidate	Rows	CER	WER	Locale violations	Decision
Whisper large-v3	15	33.33	42.04	2	bounded exploratory lane
Whisper large-v3 turbo	15	40.36	50.87	4	bounded exploratory lane
SenseVoice small	15	63.12	78.98	14	bounded exploratory lane
Qwen3-ASR 0.6B	15	64.16	81.07	15	bounded exploratory lane

Candidate models are engineering-gated evidence and are not promoted to the main ASR benchmark or selected-300 claims from this table.

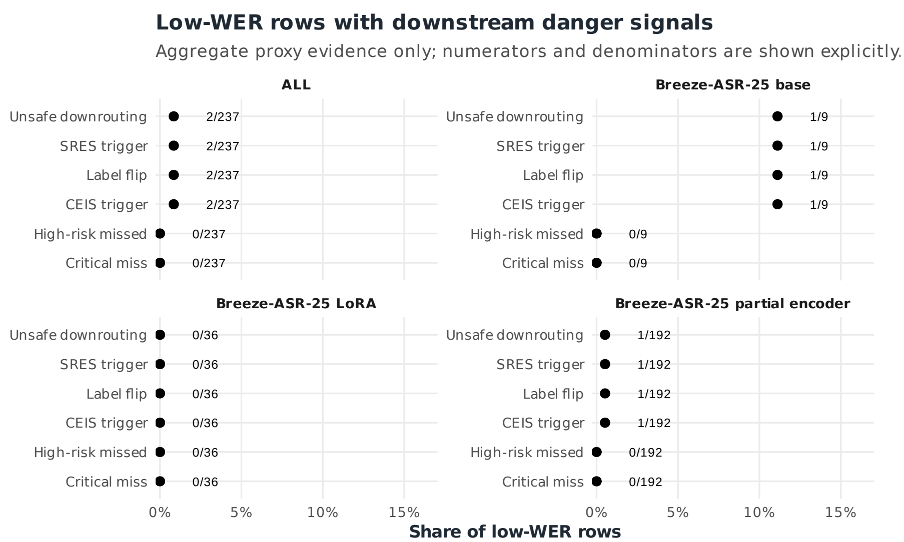

Figure 6: Low-WER danger signals under aggregate proxy evidence, with numerators and denominators shown explicitly. Source: `low_wer_danger_summary.tsv`.

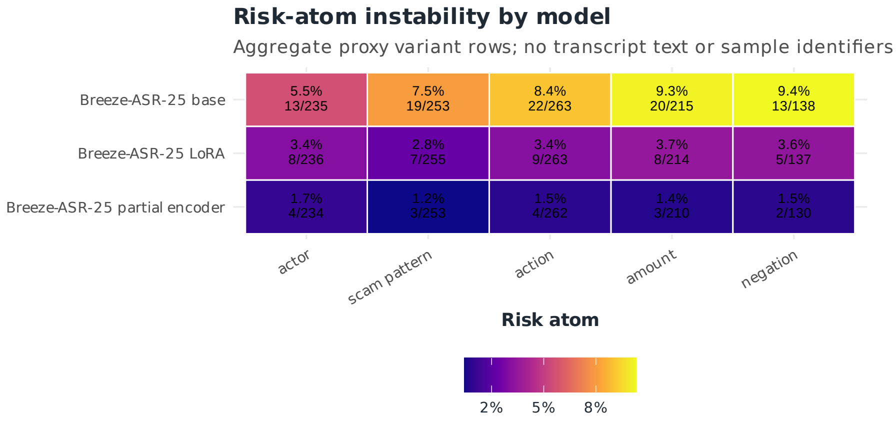

Figure 7: Risk-atom instability by model under aggregate proxy evidence. Source: `risk_atom_instability.tsv`.

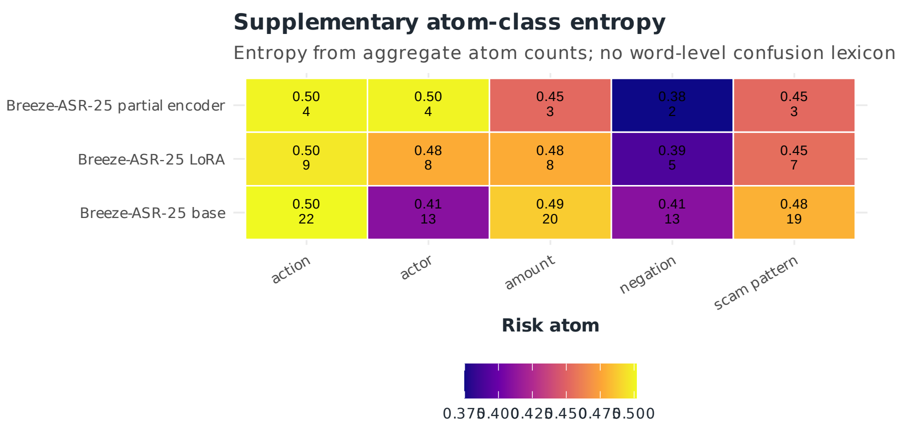

Figure 8: Atom-class entropy from aggregate instability counts. Source: `risk_atom_instability.tsv`.

Appendix Table A1. Fixed-Budget Recovery Frontier

The fixed-budget frontier is a retrospective aggregate replay over the selected 300 audio rows. It reports what remains after applying ranked conservative triggers at pre-specified 10%, 20%, 30%, and 40% row-level budgets. Tied cutoff groups are reported as best/worst-case severe remaining, and claims use the worst-case boundary.

Table 5: Final row-level fixed-budget conservative replay frontier.

Metric	Budget	Triggers	Severe	Eliminated	Unsafe	Critical	Tie best	Tie worst
SRES	10%	30	0	6	0	0	0	0
SRES	20%	60	0	6	0	0	0	0
SRES	30%	90	0	6	0	0	0	0
SRES	40%	120	0	6	0	0	0	0
CEIS	10%	30	0	6	0	0	0	0
CEIS	20%	60	0	6	0	0	0	0
CEIS	30%	90	0	6	0	0	0	0
CEIS	40%	120	0	6	0	0	0	0
Fusion	10%	30	0	6	0	0	0	0
Fusion	20%	60	0	6	0	0	0	0
Fusion	30%	90	0	6	0	0	0	0
Fusion	40%	120	0	6	0	0	0	0
Variant count	10%	30	5	1	16	1	5	6
Variant count	20%	60	2	4	8	0	2	5
Variant count	30%	90	2	4	6	0	2	5
Variant count	40%	120	2	4	6	0	2	5
CEIS capped	10%	30	0	6	0	0	0	0
CEIS capped	20%	60	0	6	0	0	0	0
CEIS capped	30%	90	0	6	0	0	0	0
CEIS capped	40%	120	0	6	0	0	0	0

Retrospective aggregate replay over 300 audio rows. Tied cutoff groups are reported as best/worst-case severe remaining; claims use the worst-case boundary.

Evidence source: `70_experiments/runs/janus_300_high_stakes_final_csl_2026_06_03/final_csl_fixed_budget_frontier_row_level.tsv`.

Operation Records

The selected-300 reviewer route is documented through aggregate-only operation records, including response closeout, response apply logs, reviewer work order, post-review sequence, operation-record audit, and consistency audit. These records provide command-level reproducibility without exposing transcript or row content.

Key records:

- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_response_closeout_summary.json`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_batch_response_apply_log.tsv`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_review_work_order_summary.json`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_post_review_sequence_summary.json`
- `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_operation_record_summary.json`

Privacy And Local-Only Boundary

The following stay local or ignored:

- raw audio;
- raw transcripts;
- selected sample IDs and audio IDs;
- transcript-bearing audit sheets and response sheets;
- reviewer notes;
- raw model hypotheses and predictions;
- transcript-bearing runtime logs;
- checkpoints, adapters, and model weights;
- local downloaded reviewer packets and zip files.

The tracked repo preserves aggregate counts, run records, validation summaries, metric tables, and paper-facing evidence matrices. This boundary lets reviewers audit the evidence chain while protecting transcript-bearing material.

Validation Gate Commands

Run these after manuscript packaging changes:

```
.venv/bin/python 80_semantic_risk_asr/scoring/audit_postdoc_roadmap_completion.py --output-j
.venv/bin/python 80_semantic_risk_asr/scoring/audit_publishable_evidence_chain.py --output-j
```

```
.venv/bin/python 80_semantic_risk_asr/scoring/audit_evidence_chain_consistency.py --output-j
```

Expected current gate state:

- roadmap completion: `roadmap_complete=true`, `blocking_gate=none`;
- publishable evidence: `publishable_ready=true`;
- consistency audit: `26/26` checks pass, `failed_checks=[]`.

Scope Control For Additional Experiments

Do not run new full-split ASR experiments now. Candidate models can move to 258-row or selected-300 only after strict Taiwan Traditional Chinese locale policy is satisfied or an isolated Gemma 4 multimodal runtime exists. Until then, the active work is manuscript drafting, results-table packaging, artifact availability, citation completion, and submission readiness.

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